SecurePsych AI

Feature POC Checklist: Summarization Service

Version 1.0 | CSIS494 | March 2026

Feature	Summarization Service — Structured Document Summarization
Owner	Arpita H.
Service Port	Microservice behind API gateway on port 8000
Framework	LangChain (confirmed) + method per Research 3 findings
Scope Note	Stateless — no session persistence between requests. One document per request.
Purpose	Minimum bar for demo readiness and merge to main
Phase Gate	Phase 2: Core RAG + Summarization (Mar 11 – Mar 28)

1. What This Checklist Is

This is a PR acceptance checklist for the Summarization service as a whole—not a task list. Before this service is considered demo-ready and eligible to merge to main, every item below must pass. The feature is stateless: there is no memory or session state between requests. The scope is a single document per summarization call—up to approximately 60 pages. Items are organized by concern: pipeline correctness, output quality, performance, and gateway integration.

2. Important Scope Note: Stateless Design

This service has no persistent memory between sessions. A user uploads a document, requests a summary, receives the output, and the session ends. The service retains no information between calls. If the chosen summarization method (e.g., Refine) passes a running summary between internal LLM calls within a single request, that is in scope and expected—but user-level state does not persist. Do not add session caching or persistent context stores to this service.

3. Acceptance Criteria

#	Acceptance Criterion	Notes / Threshold
A. End-to-End Pipeline		
1	A user can submit a PDF document (or a document ID from a prior ingestion) and receive a structured summary response in a single API call.	*Summarization endpoint accepts PDF input or document reference.*
2	The summarization pipeline applies the method recommended in Arpita H.'s Research 3 findings (expected: Refine method for structured reports). The method used must be documented in the PR.	*If Research 3 recommends a different method, that method is used. Document the decision.*

3	The summary output is structured and includes, at minimum: identified diagnoses or eligibility findings, key test scores or assessment results, accommodations or recommendations, and evaluation dates where present.	*Output does not need to be perfectly formatted for POC—content coverage is the bar.*
4	The service correctly handles a 48–60 page psychoeducational evaluation document without crashing, truncating silently, or exceeding the model's effective context window in a way that corrupts the output.	*Test on the longest available sample document.*
5	If the document exceeds the single-call context limit and requires recursive processing (e.g., map-reduce or refine across chunks), each processing step completes without silent failure.	*Log or trace each LLM call in the chain.*
B. Output Quality Bar		
6	On a sample psychoeducational evaluation document, the summary captures all major section types present in the document (e.g., background, assessment results, eligibility, recommendations). Reviewable by Natoli or Dr. Josh Johnson.	*At least one sample summary reviewed by a human before merge.*
7	The summary does not introduce information that is absent from the source document. All factual claims in the summary must be traceable to content in the uploaded PDF.	*Any fabricated content is a blocker.*
8	For the Refine method (or whichever is selected): later-document references (e.g., a recommendation on page 45 referencing a score from page 12) are preserved coherently in the final summary output.	*Spot-check 2–3 cross-references in the test document.*
C. Performance		
9	Summarization of a 48-page document completes within 90 seconds on the target hardware.	*If latency exceeds 90 seconds, async processing with a progress indicator is required before merge. Record actual measured latency in PR notes.*
10	If async processing is implemented due to latency: the endpoint returns a job ID immediately, and a polling or callback mechanism delivers the completed summary. The frontend contract for this must be documented.	*Only required if sync latency > 90 seconds. Otherwise, synchronous response is preferred.*
11	LangSmith tracing is active on the summarization chain and captures per-step latency and token counts. No document content is transmitted externally.	*Verify no document text appears in LangSmith logs.*
D. Gateway Integration		
12	The Summarization service runs as an independent FastAPI microservice and is accessible only through the API gateway on port 8000. The frontend does not call the service directly.	*Architecture constraint—same as all other services.*
13	The /health endpoint for the Summarization service returns 200 OK and is reachable through the gateway.	*Health check must pass before CI/CD marks the build green.*
14	The service respects its feature flag in config.yaml. If disabled, the gateway returns a graceful 'feature unavailable' response.	*Smoke test with feature flag toggled off.*

15	Per-request temp files (intermediate chunk outputs, LLM call artifacts) are deleted after the session completes via the shared cleanup utility.	*Verify no orphaned temp files after 3 sequential test requests.*
E. Security & Compliance Baseline		
16	The service makes zero outbound network calls during a summarization request. All LLM inference routes through the local Ollama instance.	*Confirm with: curl api.openai.com must fail while the service is processing.*
17	Any intermediate summary chunks or LLM outputs written to disk during the Refine/map-reduce chain are stored within the shared encrypted storage path.	*Spot-check file write locations during a multi-chunk summarization run.*
18	All file writes must use the SECURE_STORAGE_PATH environment variable. No hardcoded paths.	*Verified by spot-checking during ingestion run.*

4. What Is Explicitly Out of Scope for This Checklist

The following are not required for Phase 2 merge and are tracked separately:

- Frontend UI for the summary display panel — tracked under Alyse B.'s frontend work.
- Jargon simplification of summary output — Phase 3 feature (separate service).
- FERPA compliance validation — blocked pending Jon T.'s research completion.
- Multi-document summarization or cumulative file comparison — future scope.
- Reading level validation on summary output — not required until jargon simplification (Phase 3).

5. How to Use This Checklist

When the Summarization service is ready for merge, Arpita H. opens a PR and pastes this checklist into the PR description. Check off each item with brief evidence (e.g., test output, LangSmith screenshot, latency measurement, human review note from Natoli or Josh). Natoli reviews the completed checklist before approving the merge. Any unchecked item needs an explicit decision: fix it, defer it with justification, or accept it as a documented known gap.